

Research Landscape Briefing: Multilingual AI Sign-Language Avatar System for DHH Students in Bangladesh

Prepared as a PhD-supervisor-style landscape analysis. Date: 11 June 2026.

TL;DR

- The hardest, most novel parts of your project — generating *correct, fluent novel signing motion* and keeping a 3D avatar *synchronized to a live lecture timeline* — are unsolved research problems, not engineering tasks. The realistic move for your July 1 hackathon is a synchronization-first, dictionary-playback system (pre-baked clips + timeline-locked scheduling), NOT neural gloss-to-motion generation.
- The single biggest evidence-based risk is that Deaf viewers reliably rate computer-synthesized avatars far below human signers: in Quandt et al. (2022, N=191) a JASigning-style avatar scored 2.62/5 on comprehension vs. 4.62 for a human and 3.79 for motion-capture. Your value-add must therefore be educational scaffolding (captions + simplified text + avatar), not avatar fidelity.
- For Bangla Sign Language there is NO continuous-signing corpus, NO motion-capture data, NO avatar system and NO standardized gloss notation — only isolated-character/word image datasets and one new 1,000-sentence gloss-pair text dataset (2025). You are building on bare ground, so scope BdSL as a research contribution, not a solved input.

Key Findings

1. **Two technology families exist, and they have diverged.** Production deployments (Signapse, CODA, Signvrse) have largely abandoned 3D cartoon avatars in favor of *photorealistic, video-based or motion-capture-driven* signers, precisely because Deaf users reject synthetic avatars. Your VRM/Three.js approach sits in the family (graphical avatars) that the field has found *least* accepted — but it is also the only family that is lightweight, browser-deployable, customizable, and feasible for a student team. Be clear-eyed about this trade-off.
2. **Gloss-to-motion has two regimes.** (a) *Dictionary playback with blending* (HamNoSys→SiGML→JASigning, or your clip dictionary) is production-ready, deterministic, and correct per-sign, but robotic and limited to its vocabulary. (b) *Neural gloss/text-to-pose generation* (Progressive Transformers, diffusion models) is a live research frontier that produces under-articulated, often non-understandable output and needs large motion datasets you do not have for BdSL. For a hackathon, (a) is the only responsible choice.
3. **Synchronization is tractable and is your real differentiator.** Word-level forced alignment (WhisperX, ~sub-100ms timestamps) plus a timeline-locked scheduler is a known, solvable

architecture. The hard sub-problems are *time-compression* (signing is slower than speech) and *seek/pause handling*, which require an explicit buffering/queue policy.

4. **Non-manual markers are linguistically essential and technically under-served.** Eyebrow raises (questions), head-shake (negation), and mouth morphemes carry grammatical meaning; omitting them changes sentence meaning. Your VRM's blendshapes can support basic NMMs, which would be a meaningful, differentiating contribution because most dictionary systems ignore them.
5. **BdSL is a genuine research gap.** This is simultaneously your biggest risk (no data/standards) and your biggest opportunity for an original contribution.

Details

PART A — Sign Language Avatar Research

The central, well-established empirical finding (this should anchor your whole project): Deaf viewers rate synthetic avatars markedly lower than humans. In Quandt, Willis, Schwenk, Weeks & Ferster (2022, *Frontiers in Psychology* 13:730917, N=191 ASL users — 83 Deaf, 34 Hard-of-Hearing, 67 hearing), a computer-synthesized avatar (the JASigning "Anna" character driven via SiGML/HamNoSys) scored **2.62/5 for comprehension (SD 1.13), 2.17/5 for naturalness, and highest on "creepiness" (3.61)**, versus a human signer (4.62 comprehension, SD 0.56) and a motion-capture avatar (3.79). The Signer main effect was large: $F(1.39, 250.96) = 239.76, p < 0.001, \eta^2 = 0.42$. Crucially, **native/fluent signers penalize synthesized motion the most** — for the CS avatar only, comprehension ratings *decreased* with increasing fluency and with earlier age of ASL acquisition; that sensitivity disappeared for the mocap avatar. Bragg et al. (2019, ASSETS, Best Paper, "Sign Language Recognition, Generation, and Translation: An Interdisciplinary Perspective," 12 authors across Microsoft Research, RIT, Gallaudet, Boston Univ., LIMSI/CNRS, Univ. of Maryland, Leiden) states bluntly that "the state-of-the-art in avatar generation is not fully automated; all parts of current pipelines currently require human intervention to generate smooth, coherent signing avatars." **Takeaway: do not over-invest in avatar realism; invest in correctness, NMMs, and educational scaffolding.**

Influential research papers (with relevance scores 1-10 for THIS project):

- **Saunders, Camgoz & Bowden, "Progressive Transformers for End-to-End Sign Language Production" (ECCV 2020).** First model to translate spoken-language text → continuous 3D sign-pose sequence end-to-end, using a novel counter-decoding technique; offers both end-to-end and gloss-intermediary configurations; evaluated on PHOENIX14T. *Problem solved:* continuous (not isolated) SLP. *Limitation:* regression-to-the-mean produces under-articulated, robotic motion; manual-only in base form. **Relevance: 7/10** — the canonical SLP paper you must understand conceptually, but you cannot train it for BdSL (no data). Open-source (BenSaunders27 GitHub). ([ResearchGate](#)) ([arXiv](#))

- **Saunders et al., "Adversarial Training for Multi-Channel Sign Language Production" (BMVC 2020) and "Continuous 3D Multi-Channel SLP via Progressive Transformers and MDNs" (IJCV 2021).** Add an adversarial discriminator + mixture density networks to produce *non-manual* features (face, mouthing) and reduce regression-to-mean. **Relevance: 6/10** — shows why NMMs and multi-channel output matter, and demonstrates a discriminator measurably improves Gloss2Pose comprehension.
- **Saunders et al., "Everybody Sign Now" / SignGAN (2020) and "Signing at Scale" (CVPR 2022).** Pivot from graphical avatars to *photorealistic* video generation, explicitly because "prior approaches based on graphical avatars have proven unpopular, whereas recent neural SLP works that produce skeleton pose sequences have been shown to be not understandable to Deaf viewers." **Relevance: 5/10** — important context (why industry went photorealistic) but out of scope for a browser VRM. (arxiv)
- **Baltatzis et al., "Neural Sign Actors: A Diffusion Model for 3D Sign Language Production from Text" (CVPR 2024).** Diffusion-based SLP trained on a curated large-scale dataset of 4D signing avatars; generates 3D avatar sequences; outperforms prior SLP. *Limitation:* heavy, data-hungry, not real-time/browser. **Relevance: 6/10** — the current 3D SOTA; aspirational, shows where the field is going.
- **Fang et al., "SignLLM: Sign Language Production Large Language Models" (2024) and "SignDiff" (2023/2025).** LLM- and diffusion-based ASL production; report SOTA on How2Sign/PHOENIX14T. **Relevance: 4/10** — relevant to your LLM gloss step conceptually.
- **Shalev-Arkushin, Moryossef & Fried, "Ham2Pose: Animating Sign Language Notation into Pose Sequences" (CVPR 2023).** First method to animate HamNoSys notation → pose sequences, *language-invariant by design*, trained with weak supervision; introduces nDTW (normalized DTW) as a distance metric. **Relevance: 8/10** — this is the most directly transferable research idea for you: notation→motion that is not tied to one sign language, exactly the generalization BdSL needs. Open-source (rotem-shalev GitHub).
- **Alibaba XR Lab, "A Speech-driven Sign Language Avatar Animation System for Hearing Impaired Applications" (IJCAI 2022) and companion "Text/Speech-Driven Full-Body Animation" (IJCAI 2022).** Production-oriented speech→avatar full-body synthesis pipeline (face + body skinned and rendered to a video stream). **Relevance: 6/10** — architecturally the closest published analog to your goal (speech in, avatar out).
- **Rastgoo, Kiani, Escalera et al., "Sign Language Production: A Review" (CVPR-W 2021) and "A survey on recent advances in SLP" (ESWA 2023).** **Relevance: 7/10** — best single survey entry-point.
- **Moryossef, "Real-Time Multilingual Sign Language Processing" (PhD thesis, Bar-Ilan, 2023) and the associated sign.mt application.** **Relevance: 8/10** — a real-time, multilingual, open-source-minded systems perspective extremely close to your framing; the survey chapter covers JASigning, CWASA, PAULA, and the ViSiCAST lineage.

Systems (purpose / strengths / weaknesses / similarity to your project):

- **JASigning (University of East Anglia).** *Purpose:* avatar renders SiGML (XML based on HamNoSys notation) into real-time signing; runs cross-platform (Java). *Strengths:* linguistically rigorous, notation-driven, supports manual + non-manual, multiple avatars (Anna, Marc, etc.), decades of development (ViSiCAST/eSIGN/DictaSign lineage), free for research. *Weaknesses:* dated rendering, robotic, low Deaf-acceptance scores, not a modern web/Three.js stack. *Similarity:* HIGH — it is the closest conceptual ancestor of a gloss/notation→avatar pipeline. Study its SiGML→animation architecture closely.
- **SignON (EU Horizon 2020, 2021–2023).** *Purpose:* multi-modal translation between several European signed/spoken languages with avatar (EVA) output. Confirmed details: Horizon 2020 Grant Agreement No. 101017255, €5.6M, 17 partners, led by Prof. Andy Way (Dublin City University, coordinator) and Dr. Dimitar Shterionov (Tilburg University, scientific lead); commenced 1 Jan 2021; targets ISL, BSL, NGT, VGT (Flemish SL) and LSE (Spanish SL). *Strengths:* co-creation with Deaf communities, open framework, multilingual interlingua. *Weaknesses:* the EVA avatar was "early stage"; project ended without a production avatar. *Similarity:* MEDIUM-HIGH — your end-to-end pipeline mirrors theirs; read their "Sign Language Technology: Do's and Don'ts" white paper for ethics/co-creation lessons.
- **Signapse (UK, spun out of Univ. of Surrey CVSSP; £2M seed in 2024 — £1.5M investors + £500k Innovate UK).** *Purpose:* photorealistic AI BSL/ASL via a two-step Text→Gloss then Gloss→Video pipeline, stitching qualified video clips. *Strengths:* photorealistic, deployed at scale — Signapse's AI generates 5,000 BSL train announcements daily, "with a record peak of 8,862 daily train announcements in BSL recorded in May 2024" (Surrey Research Park), serving ~87,000 UK BSL Deaf users; Deaf-led QC. *Weaknesses:* domain-limited, leans on Sign-Supported English word order, not 3D/avatar, not open. *Similarity:* MEDIUM — same Text→Gloss→render architecture, but video not avatar.
- **CODA (Israel) and Signvrse (Kenya).** Per Savatia (Signvrse CEO) in TechCabal (21 Aug 2025), the flagship "Terp360 draws on a dataset of over 20,000 professionally captured sign sequences" (60% urban, 40% rural signers), uses motion-capture of Deaf signers + LLM semantics + real-time 3D skeletal data driving a "hyper-realistic" avatar for Kenyan Sign Language with explicit non-manual marker support, and "achieved a 35% reduction in processing time by pre-computing common sign combinations." Their verbatim design principle is directly instructive for your sync layer: "It's better to show a slightly simpler sign immediately than a perfect one after a long delay." *Similarity:* HIGH (Signvrse) — architecturally the closest current startup to your vision.
- **Hand Talk (Brazil, "Hugo"/"Maya" avatars).** *Purpose:* app translating Portuguese/English → Libras/ASL/BSL via 3D cartoon avatar; AI "trained with over 38,000 validated words." *Strengths:* huge adoption and recognition — won "Best Social App of the World in 2013, handed by UN at the World Summit Mobile Award, in Abu-Dhabi." *Weaknesses:* user reviews report heavy fingerspelling and weak grammar; a verbatim App Store/Google Play review states, "The avatar can't even sign basic colors or days of the week, all it does is fingerspell... when it does occasionally use the proper sign for a word, it still isn't very

accurate." *Similarity*: HIGH — closest consumer analog to your 3D-cartoon-avatar approach; instructive about its failure modes (fingerspelling fallback is the cardinal sin to avoid).

- **SiMAX (Austria)**. *Purpose*: 3D avatar text→sign with human-translator-in-the-loop. *Strengths*: recognized (World Summit Award), customizable. *Weaknesses*: not real-time/automatic, commercial. *Similarity*: MEDIUM.
- **SignAvatar/TransportSign (Serbia)**. *Speech*→sign avatar for airport/train announcements; ~3-4s latency; ~6-week per-deployment workflow. *Similarity*: MEDIUM — useful latency benchmark.

PART B — Gloss-to-Avatar Research

The notation stack is the production-ready path and you should adopt its concepts:

- **HamNoSys (Hamburg Notation System)**: a phonetic transcription describing handshape, location, orientation, movement (manual) plus non-manual features. Universal across sign languages. Complex to author by hand.
- **SiGML (Signing Gesture Markup Language)**: XML encoding of HamNoSys; the input format consumed by JASigning to drive avatar animation in real time.
- **SignWriting**: a visual, 2D phonological script (used in Moryossef/Jiang machine-translation work).
- **Gloss**: sign-to-word semantic identifiers in spoken-language order-adjusted form (what your LLM produces).

How existing systems solve gloss→motion:

1. **Notation-driven procedural animation (HamNoSys→SiGML→JASigning)**: production-ready, deterministic, linguistically grounded, generalizes across languages. Best fit for educational content where *correctness* matters. *Downside*: robotic, requires authoring HamNoSys per sign.
2. **Dictionary clip playback + blending (your current approach)**: production-ready and simplest; quality depends on clip library and blend quality. Best for fixed educational vocabularies. *Downside*: vocabulary-bounded, co-articulation/transition artifacts.
3. **Neural gloss/text→pose (Progressive Transformers, Ham2Pose, diffusion)**: research-grade; needs large pose/motion datasets (PHOENIX14T, How2Sign); output often under-articulated. NOT production-ready, NOT feasible for BdSL now.
4. **General human-motion text-to-motion models (MDM — Motion Diffusion Model, Tevet et al. 2022/2023; T2M-GPT; MotionGPT)**: SOTA for *general* motion on HumanML3D/KIT, but sign language requires precise hand/finger articulation and NMMs that these models do not capture; not trained on sign data. **Relevance: low-medium** — useful background on motion generation, not directly usable.

Which methods are production-ready vs. research, and which suit educational content:

Methods 1 and 2 are production-ready and the correct choice for educational content, where per-sign correctness is non-negotiable. Methods 3 and 4 are research problems unsuitable for a deadline-bound, low-resource (BdSL) educational deployment.

Best datasets (for your literature framing; none are BdSL):

- **RWTH-PHOENIX-Weather-2014T** (German SL, ~11h, weather domain, 7,096/519/642 train/dev/test segments, ~1,085 gloss vocab) — the canonical SLP/SLT benchmark, though narrow.
- **How2Sign** (ASL, ~79-80h, 16K vocab, 31,128/1,741/2,322 segments; multimodal: video+pose+depth+speech) — best multimodal SLP resource.
- **CSL-Daily** (Chinese SL, ~23h, 2K vocab, 18,401/1,077/1,176 segments), **BOBSL** (BSL, ~1,400h, ~78K vocab, TV, 993K train segments) — large-scale.
- **YouTube-ASL** (ASL, ~984h, 60K vocab, web) — largest open ASL.

Best open-source implementations to study: Progressive Transformers (BenSaunders27), Ham2Pose (rotem-shalev), MDM (GuyTevet), three-vrm + vladmandic/human-three-vrm for VRM animation.

PART C — Real-Time Synchronization Research

This is your most tractable hard problem and your best opportunity to ship something that works.

Forced alignment / word-level timestamps: **WhisperX** (Bain, Huh, Han & Zisserman, "WhisperX: Time-Accurate Speech Transcription of Long-Form Audio," Interspeech 2023, pp. 4489-4493, doi:10.21437/Interspeech.2023-78) is the key tool — VAD pre-segmentation + Whisper transcription + wav2vec2 forced phoneme alignment yields word-level timestamps with reported sub-100ms accuracy (vs. ~500ms-1s drift in vanilla Whisper) and 60-70× real-time speed with large-v2 via batched inference (requiring <8GB GPU memory with beam_size=5, per the m-bain/whisperX repo). (Caveat: GitHub issues report word-timestamp accuracy regressions in some recent versions versus Montreal Forced Aligner, which is more accurate but slower and not streaming.) For *live mic* input you need a streaming ASR + incremental alignment, which is materially harder than file-based alignment.

The core architectural problem is that *sign language is slower than speech*: a faithful signed rendering of a fast lecture cannot stay frame-locked without falling behind. Practical systems therefore choose among:

1. **Timeline-locked scheduling (most practical):** for pre-recorded YouTube video, get word timestamps, map gloss to clips, and schedule clips against the media `currentTime`. Deterministic, seek/pause-friendly. **Recommended for the hackathon.**
2. **Streaming buffer / queue (for live mic):** transcribe → gloss → enqueue signs; play from a buffer that absorbs jitter. Signvrse's "show a slightly simpler sign immediately than a perfect

one after a long delay" plus pre-computing common combinations is the canonical industrial pattern.

3. **Event-driven (least timeline-aware):** play whenever text arrives; simplest but drifts and breaks on seek.

How virtual interpreters/broadcast anchors maintain timing: broadcast/transport deployments (Signapse, SignAvatar) accept a fixed latency offset (~3-4s) and run a buffered queue rather than attempting zero-latency sync — they trade immediacy for stability, which is the right trade for lectures too.

What is solved: word-level alignment for recorded audio; clip scheduling; seek/pause for *recorded* media. **What remains hard/unsolved:** real-time alignment for live streaming audio; graceful time-compression when signing falls behind speech (sign summarization/skipping is an open research question); maintaining sync across variable-length generated motion.

Ranking, most practical → most experimental:

1. Timeline-locked clip scheduling against `video.currentTime` (recorded) — **production-ready**.
2. Streaming buffer with catch-up/skip policy (live) — **engineering-feasible, needs design**.
3. Neural generation synced to timeline — **research-grade, avoid for now**.

PART D — Facial Expressions and Non-Manual Markers (NMMs)

NMMs are grammatically obligatory, not decorative. Established linguistics:

- **Eyebrows:** raised = yes/no questions and topic/conditional marking; furrowed = wh-questions.
- **Head:** shake = negation (the grammatical core of negation, often without a separate NOT sign); nod = affirmation; tilt = topicalization/conditionals.
- **Mouth morphemes:** distinct mouth shapes adding adverbial/adjectival meaning (size, manner, intensity) — distinct from mouthing spoken words.
- **Eye gaze:** marks directionality, role shift, agreement.

Which components matter most for comprehension: eyebrows (question/topic marking) and head-shake (negation) carry the most sentence-level grammatical load and are the highest-value targets; mouth morphemes and mouthings disambiguate otherwise-identical manual signs.

What is solved: rendering individual blendshape-based facial expressions on a rigged avatar is straightforward (VRM supports ARKit-style 52-blendshape sets; eyebrow/mouth/eye morphs).

What remains hard: *timing and co-articulation* — synchronizing NMM onset/offset with the correct span of manual signs, and generating them automatically from text. The Wasserroth et al. (2025, IVA Adjunct, N=9, DFKI AVASAG HoloLens-2 avatar) study found "less than 50% of the individual sentence parts were understood," attributing failure "mostly due to lack of mouthings

and facial expressions, and difficulty of distinction between similar hand shapes" (e.g., the manual sign for "Hanover" identical to "blue" without mouthing). Saunders' multi-channel/adversarial work directly targets NMM generation. **Opportunity:** even rule-based NMMs (eyebrows up on "?", head-shake on negation, mapped from your LLM's syntactic analysis) would put you ahead of most dictionary-playback systems.

PART E — 3D Avatar Models

Model	Browser/Three.js	Finger rigging	Face blendshapes	Lightweight	Licensing	Sign-language fit
VRM (VRoid/three- vrn) — your current	Native (glTF 2.0 + three-vrn)	Yes (humanoid hand bones; your 124-bone rig is adequate)	Yes (ARKit-style + VRM expressions)	Yes (your 12MB/36k-poly model is ideal)	Open (VRM, per-model license)	Good for prototype; finger detail may need a richer hand rig
Ready Player Me	glTF/GLB (web-friendly)	Limited (basic hand bones; not signing-grade)	Yes (ARKit 52, Oculus visemes)	Yes	Free w/ license terms	Weak — insufficient finger articulation
Unreal MetaHuman	No (Unreal-bound; heavy)	Excellent	Excellent	No (very heavy)	Epic license	High fidelity but NOT browser-deployable
Blender custom	Export to glTF	Fully customizable (you control bone count)	Customizable shape keys	Tunable	Open	Best control, highest effort

Verdict: Stay with VRM/three-vrn. It is the only option meeting all your constraints (browser, lightweight, finger + face rigging, open). VRM's standardized humanoid bones (T/A-pose reference, standard bone names) enable animation retargeting (e.g., from Mixamo or motion capture) without manual bone mapping — useful if you later acquire BdSL mocap. The main limitation is finger articulation fidelity; verify your hand rig has per-finger joints sufficient for

distinct handshapes. MetaHuman is out (not web). Blender-custom is a later upgrade path if you need a richer hand/face rig. On performance, ready-made/lightweight avatars are explicitly the recommended choice for web/casual contexts; the cinematic realism of MetaHuman is irrelevant when the bottleneck is signing correctness, not polygon count.

PART F — Bangla Sign Language (BdSL)

What exists:

- **Ishara-Lipi** (Islam, Mousumi, Jessan, Rabby & Hossain, ICBSLP 2018): first complete isolated-character BdSL dataset, 36 characters (30 consonants, 6 vowels), 50 sets, ~1,800 images after preprocessing. Image classification only.
- **Ishara-Bochon, BDSL-49, BdSL-36/40, BAUST Lipi** (18,000 images, 36 symbols), **BdSLW60** (IUT; ~9,000 samples, 60 words, MediaPipe + attention BiLSTM, ~75% test accuracy), **MV-BSL / MVBSLW50** (multi-view, 9,204 videos, 107 sentences + 350 words, 40 signers) — all *recognition* datasets (sign→text), isolated or small-vocabulary.
- **Bangla-SGP** (arXiv 2511.08507, 2025): the first Bangla sentence→gloss pair dataset — 1,000 human-annotated pairs + ~3,000 RAG-augmented pairs, annotated by a single professional signer (signer bias explicitly acknowledged). This is the most relevant resource for your gloss step.

Active groups/universities: BUET, Dhaka University, CUET, BRAC University, NSU, Islamic University of Technology (IUT, BdSLW60), Bangladesh Army University of Science and Technology (BAUST). Research is dominated by recognition, not production.

What does NOT exist for BdSL (the gaps — and your opportunity):

- No continuous-signing video corpus at scale.
- No motion-capture / 3D-pose data.
- No avatar or sign-production system.
- No standardized, community-ratified gloss notation or HamNoSys transcription set for BdSL.
- No published educational-effectiveness study of a BdSL avatar.

Implication: you cannot train a neural BdSL production model. Your only viable path is a curated clip/notation dictionary + LLM gloss + sync, with the BdSL gloss conventions themselves treated as a research contribution (build on Bangla-SGP).

PART G — Problem Maturity Table

Challenge	Current State	Difficulty	Research Maturity	Risk Level
Speech-to-text for Bengali	Works (Whisper supports Bengali) but lower accuracy than English	Medium	Mature (general ASR)	Medium
Text simplification	LLMs do this well	Low-Medium	Mature	Low
Gloss generation via LLM	Feasible; quality unverified for BdSL; Bangla-SGP exists	Medium	Emerging	Medium-High
Sign dictionary lookup	Trivial once dictionary exists	Low	Mature	Low
Pre-baked clip playback	Solved	Low	Mature	Low
Smooth animation blending	Solved-ish (transition artifacts)	Medium	Mature in graphics	Medium
Gloss→novel-motion generation	Research-grade, under-articulated	Very High	Active research	Very High (avoid for hackathon)
Non-manual markers	Rendering solved; auto-timing hard	High	Active research	High
Real-time synchronization (recorded)	Solvable (forced alignment + scheduler)	Medium	Mature components	Medium
Real-time sync (live mic)	Harder (streaming alignment, time-compression)	High	Emerging	High
Seek/pause handling	Solvable with timeline-locked design	Medium	Engineering	Medium
BdSL linguistic correctness	Largely unestablished; no standard	High	Nascent	High
Educational-effectiveness	Instruments exist (UEQ-S, NASA-TLX, TAM/TAMSA,	Medium	Mature methods	Medium

Challenge	Current State	Difficulty	Research Maturity	Risk Level
measurement	comprehension-question accuracy); not applied to BdSL avatars			

PART H — Research Roadmap (ranked, most → least critical)

For July 1, 2026 hackathon (build these):

1. **Real-time synchronization architecture (recorded YouTube first).** *Why it matters:* it's the spine of the demo and your most defensible contribution. *Existing solutions:* WhisperX word timestamps + timeline-locked clip scheduler against `video.currentTime`. *Remaining gaps:* time-compression and seek/pause policy. *Direction:* implement a buffer + catch-up/skip rule; demo seek/pause robustness explicitly.
2. **Gloss-to-clip dictionary playback with blending.** *Why:* the only production-ready way to render correct signs. *Existing:* your 30–80 clip dictionary + three-vm. *Gap:* vocabulary coverage, transition smoothness. *Direction:* expand dictionary for one lecture domain; implement cross-fade/IK blending.
3. **LLM gloss generation tuned to BdSL conventions.** *Why:* drives everything downstream. *Existing:* Groq LLM + Bangla-SGP dataset. *Gap:* BdSL grammar correctness. *Direction:* few-shot prompt with Bangla-SGP examples; validate with a Deaf signer.
4. **Basic rule-based non-manual markers.** *Why:* cheap, high-impact differentiator. *Direction:* eyebrow/head-shake morphs triggered by LLM syntax tags.

Longer-term (post-hackathon research): 5. BdSL gloss notation + continuous corpus (the foundational contribution). 6. Notation→motion à la Ham2Pose for vocabulary generalization beyond the clip dictionary. 7. Motion capture of Deaf BdSL signers (Signvrse model) to raise acceptance toward mocap-level. 8. Educational-effectiveness study with DHH students using validated instruments.

PART I — Paper Reading Priority

Tier 1 (read immediately):

1. **Moryossef, "Real-Time Multilingual Sign Language Processing" (PhD thesis, 2023)** — systems-level, real-time, multilingual; closest to your framing; covers the whole pipeline and avatar lineage. *Gain:* the end-to-end architecture mental model.
2. **Quandt et al. (2022), *Frontiers in Psychology* 13:730917** — the avatar-acceptance evidence base. *Gain:* sober expectations; why to prioritize scaffolding over fidelity; the fluency-sensitivity result that should shape your evaluation plan.

3. **Bragg et al. (2019), ASSETS, "SLR, Generation & Translation: An Interdisciplinary Perspective"** — the field map + Deaf-community ethics. *Gain*: what's solved vs. not; the co-design imperative.
4. **Shalev-Arkushin, Moryossef & Fried, "Ham2Pose" (CVPR 2023)** — notation→pose, language-invariant. *Gain*: the most transferable technical recipe for BdSL generalization.
5. **Saunders, Camgoz & Bowden, "Progressive Transformers for SLP" (ECCV 2020)** — canonical neural SLP. *Gain*: understand the SOTA you are *not* using and why.
6. **Bain et al., "WhisperX" (Interspeech 2023)** — your alignment engine. *Gain*: how to get word-level timestamps for sync.
7. **Bangla-SGP paper (arXiv 2511.08507, 2025)** — the only BdSL gloss resource. *Gain*: BdSL gloss conventions + augmentation method.

Tier 2 (after Tier 1):

- Rastgoo et al. SLP survey (ESWA 2023) — comprehensive landscape.
- Saunders et al. "Adversarial Training for Multi-Channel SLP" (BMVC 2020) — NMM generation.
- Baltatzis et al. "Neural Sign Actors" (CVPR 2024) — 3D diffusion SOTA.
- Kipp, Heloir & Nguyen avatar comprehensibility studies (IVA/ASSETS 2011) — evaluation methodology ("delta testing"); and Kacorri et al. (ASSETS 2015 / ACM TACCESS 2017) on demographic predictors of acceptance.
- Ebling & Glauert (2016, *Univ. Access Inf. Soc.* 15(4):577–587) — JASigning train-announcement evaluation; concrete avatar-design pitfalls (clothing color, eyegaze, fingerspelling speed, manual/non-manual coordination).
- Marschark et al. (2006, *JDSDE* 11(4):421–437) — DHH classroom learning: real-time text alone outperformed sign interpreting and both combined, yet all conditions left deaf students below hearing peers; students overestimate their own comprehension.

Tier 3 (reference):

- MDM (Tevet et al. 2022), T2M-GPT, MotionGPT — general motion generation background.
- SignLLM, SignDiff, SignGen — recent neural production.
- How2Sign, PHOENIX14T, BOBSL dataset papers — for data framing.
- SignON white paper "Sign Language Technology: Do's and Don'ts."
- Stinson, Elliot & Kelly (2009/2017) — speech-to-text vs. interpreting retention and perceptions (grade-level moderates the C-Print advantage).

Recommendations

Stage 0 (now → July 1): ship a synchronization-first, recorded-video MVP.

- Lock scope to **recorded YouTube + one lecture domain** in Bangla *or* English (not both at full quality). Live mic is a stretch goal, not core.
- Architecture: WhisperX word timestamps → LLM gloss (few-shot with Bangla-SGP) → clip-dictionary lookup → timeline-locked VRM playback with blending → captions + simplified text shown alongside.
- Implement **explicit seek/pause handling** and a **catch-up/skip buffer** — demo these; they are what separates a real system from a toy.
- Add **rule-based NMMs** (eyebrows on questions, head-shake on negation) as a visible differentiator.
- **Recruit at least one Deaf BdSL signer as a collaborator/evaluator now.** Every system that succeeded did co-design (SignON, Signapse, Signvrse); every one that failed skipped it.

Benchmarks/thresholds that change the plan:

- If word-timestamp drift exceeds ~250ms in practice → add a buffering offset; do not attempt frame-perfect lip-style sync.
- If clip-dictionary coverage of your target lecture is <80% of content words → narrow the domain further rather than fingerspelling everything (Hand Talk's central user complaint).
- If a Deaf evaluator rates comprehension below ~3/5 → the bottleneck is sign correctness/NMMs, not avatar looks; do not spend time on rendering polish.
- Only pursue **neural gloss-to-motion** if/when you have a continuous BdSL motion corpus (you don't yet) — otherwise it will produce non-understandable output.

Stage 1 (post-hackathon): build the BdSL gloss-notation standard + continuous corpus; this is the publishable contribution and the precondition for everything more ambitious.

Stage 2: motion capture of Deaf signers (to lift acceptance toward mocap level) and a formal educational-effectiveness study (using UEQ-S/NASA-TLX/comprehension-question instruments).

Caveats

- **Avatar-acceptance figures** (Quandt et al.) are *subjective self-ratings* of ease of understanding, not objective comprehension-question scores; the direction (synthetic << mocap << human) is robust and replicated, but exact numbers are study-specific. The hardest objective evidence comes from Kipp's "delta testing" and ISIT-style intelligibility methods.
- **WhisperX timestamp accuracy** varies by version and language; Bengali alignment models are less mature than English, and some recent WhisperX versions show word-timestamp regressions vs. Montreal Forced Aligner. Validate empirically.

- **Several startup claims** (Signvrse's 35% speedup and 20,000-sequence dataset, CODA's "near-instantaneous," SignAvatar's 3–4s latency, Signapse's 8,862-announcement peak) are company/park self-reports, not peer-reviewed — treat as directional.
- **Bangla-SGP** relies on a single annotator (acknowledged signer bias) and is small (1,000 human pairs); treat as a seed, not ground truth.
- **Production vs. research:** Signapse/CODA/Signvrse are products with non-public methods; published SLP models (Progressive Transformers, diffusion) are research prototypes that do not reach Deaf-acceptable quality. Your project lives between these and should be honest that it is a *prototype*.
- The small-sample avatar-comprehension studies (Wasserroth et al. 2025, N=9; AR-caption studies, N=24) are illustrative, not confirmatory; the former ran no significance tests on comprehension.
- The dominant, well-established conclusion across the literature is that **graphical avatars under-perform with Deaf audiences**; your contribution should be framed as *educational scaffolding for DHH students* (captions + simplified text + synchronized signing support), not as a replacement for human interpreters.